OPTIMA ACADEMY ONLINE · FACILITATOR GUIDE

Teacher Manual

8th Grade Pre-Algebra

Full-year Grade 8 course · 12 modules across 4 quarters · aligned to the Florida B.E.S.T. Standards for Mathematics · 2026-2027

⚠ **Facilitator copy.** This manual contains grading guidance, answer-checking notes, and teacher-only planning material. **Do not publish it to students.** Keep it in an unpublished teacher module or outside Canvas entirely.

How to use this manual. Read Sections 1 and 2 first - they tell you what you are facilitating and, if you teach live, how to turn the self-paced course into a two-day-a-week class. After that, treat the manual as a reference: pull up the grading, rubric, or standards section when you need it. Everything here is drawn from the built course (the 12 formatted modules, the lesson maps, and the Canvas export). Where a school-level setting still needs confirming, you will see a **VERIFY** tag.

Contents

1 · How this course works
2 · Running this course live (2 days/week)
3 · Interactives, VR & completion items
4 · AI & VR (student-facing)
5 · Using Ignite AI to support students
6 · Grading categories & gradebook setup
7 · Rubrics for teacher-graded items
8 · Assessment & activity catalog
9 · Coach, not catch
10 · Weekly announcement scripts
11 · Teacher workflow & workload map
12 · Standards coverage map
13 · What makes this course classical
14 · Sensitive content

1 · How this course works

A typical lesson

Each module is built from 2-5 *overarching lessons* (e.g., Lesson 1.01), and each of those is broken into 3-5 *mini-lessons* (1.01a, 1.01b, ...). A mini-lesson page opens with the branded navy header, then walks the concept through worked examples and short interactive checks, and ends by linking its **Independent**

Practice - 4 to 10 problems with unlimited attempts and per-question feedback (right/wrong plus a hint, then retry). Most lessons also carry a **VR Lesson Resource** that reinforces the concept immersively. The graded touchpoint a student reaches from a lesson page is almost always the Independent Practice.

A typical week (On-Demand)

Modules are paced at roughly 15 school days each. A normal self-paced week looks like this:

Segment	What the student does	Graded?	Attempts
Mini-lessons (2-4)	Read/interact with lesson pages; work examples	No	-
Independent Practice (per mini-lesson)	4-10 problems with hints and retry	Yes (practice)	Unlimited
Mid-Module Check	Short check partway through the module	Yes (practice)	Unlimited
VR Lesson Resource	Immersive activity tied to the concept	Completion	-
Summative Review	Reviews the whole module; unlimited attempts	Completion	Unlimited
Summative Assessment	Graded for correctness at module end	Yes (major)	2

Because practice items allow unlimited attempts with feedback, most of a student's week is self-correcting. Your graded attention concentrates on the **Summative Assessments** and on any VR project or alternative assignment.

Major assessments

Every module ends with a **Summative Review** (completion, unlimited attempts - a study tool) followed by a **Summative Assessment** (scored for correctness, two attempts, answers not revealed but students can check as they go). These twelve assessments are the backbone of the grade. Mid-Module Checks (one to two per module) are the early-warning signal that a student is drifting before the summative.

2 · Running this course live: a 2-day-a-week weekly model

The course is authored asynchronously and works unchanged for On-Demand scholars. If you teach a **live section that meets twice a week**, you do not rewrite anything - you use the same lessons, practice, and assessments and wrap two live meetings around them. The rhythm below turns one module (~15 async days) into roughly **3 weeks / 6 live sessions**, so the full 12-module course fits a ~36-week year with room for review.

> **The core move:** live time is for the thinking students cannot get from a screen - launching a concept, working problems together, and clearing misconceptions. Push the delivery (reading lesson pages, watching examples) and the drill (Independent Practice) to the async time between meetings. Come to class to *do math*, not to re-read the page.

The reusable weekly template

	Before class (async)	In the live session (~50 min)	After class (async)
Day 1 Launch & teach	Students preview the week's first 1-2 mini-lesson pages and the VR resource.	**10** warm-up / Mid-Module Check item as a "do now" · **25** guided instruction on the new concept with 2-3 problems worked together · **15** students start that mini-lesson's Independent Practice while you circulate.	Finish Independent Practice for the mini-lessons taught (unlimited attempts).
Day 2 Practice & extend	Attempt the next mini-lesson's practice; flag one problem you got stuck on.	**10** review the flagged/most-missed practice problems (pull from Canvas item analysis) · **25** partner or small-group problem set / VR activity debrief · **15** exit-ticket problem and a preview of next week.	Complete remaining practice + the Summative Review when the module nears its end.

How the pieces map to live time

Async element	In a live 2-day week
Mini-lesson pages	Assigned as pre-reading; you re-teach only the hard part live, not the whole page.
Independent Practice	Started in class (you circulate and catch errors live), finished as homework. This is your best live diagnostic.
Mid-Module Check	Use as a live warm-up or exit ticket; it tells you what to reteach on Day 2.
VR Lesson Resource	Do individually async; spend 10 minutes of a Day 2 session on a shared debrief or gallery walk of what students built.
Summative Review	Assign the class period before the assessment; run a live Q&A / game review over it.
Summative Assessment	Keep async and individual (two attempts). Do not spend a live session on it; use class time to prepare, not to sit the test.

Pacing at a glance

A dependable default is **one overarching lesson per week** (two live sessions), so a 3-lesson module takes ~3 weeks and a 5-lesson module (Module 6) takes ~5. Modules 4 and 6 are the densest - build in an extra

review week there. Keep the Summative Assessment window open for a few days after the last live session so students who need a second attempt have time.

> **If you lose a session** (holiday, VR outage): protect the Day-1 launch, and move the Day-2 practice fully async for that week rather than skipping the new concept. The async course is always there as the backstop - no student falls behind because a live meeting was missed.

3 · Interactives, VR & completion items

Three things new facilitators most often misjudge:

- **Independent Practice is self-correcting, not busywork.** Every practice item gives immediate right/wrong feedback with a hint and unlimited retries. A student who skips practice does not show up as a dramatic missing grade - it surfaces later as a weak Summative Assessment. Watch the practice completion and Mid-Module Checks as the early signal.
- **Summative Reviews are completion-scored study tools.** They carry unlimited attempts and exist so students can rehearse before the graded assessment. Score them for completion, not correctness.
- **VR Lesson Resources cost you almost nothing to grade** but pay off if you debrief them. They produce reinforcement, not a submission to mark. In a live section, a 10-minute shared debrief turns them into real discussion.

> There are no intro/outro *survey-quiz wrappers* in this course as built. If you want the cohort-triage benefit those provide in other Optima courses, the closest existing signal here is the **Mid-Module Check + Canvas item analysis** - use them the same way (spot the concept a cluster of students missed, then reteach it).

4 · AI & VR in this course (student-facing)

AI. The syllabus sets the posture: AI is a welcome support for understanding (explain-it-differently, check-my-thinking, brainstorm), but it may never do the reasoning an assignment asks for, and students disclose when they use it. In math the honest line is simple - a tool can check an answer, but the student must be able to reach and justify it without one. You are not required to add any AI activity; just hold that line when it comes up.

VR. Every lesson has a VR component, and several modules include a VR project. Students enter the immersive activity, do the task, and (for projects) capture their work; you review the capture or run a shared debrief. Describe VR by what students *do* in it - the platform is being replaced, so avoid naming a specific one. If a student cannot get into VR, the alternative assignment (see the module's lesson map) keeps them on pace.

5 · Using Ignite AI to support students (teacher-facing)

> **VERIFY** Ignite AI's exact features vary by Canvas release and by what Optima has enabled. Confirm the

current feature set on your instance before relying on a specific button.

Ignite AI is *your* tool for making a large asynchronous roster feel personal. Used well it saves the most time exactly where this course concentrates teacher effort - feedback and outreach. In functional terms, use it to:

- **Draft individualized feedback** on a weak Summative Assessment or a VR project, which you then edit and own.
- **Summarize Canvas item analysis** for a module so you can see the one or two problems the cohort missed and reteach them.
- **Surface students who need a nudge** - stalled practice, a missed Mid-Module Check, a declining trend - and draft a warm check-in message.
- **Generate variant practice problems** for a student who needs another rep on a skill (e.g., more Laws-of-Exponents drill).
- **Communicate with families** in clear, kind summaries, translating when helpful.

Standing rule: AI drafts, the teacher decides. You read, edit, and own everything sent to a student or family. AI is never used to surveil, rank, or auto-penalize.

6 · Grading categories & gradebook setup

VERIFY The course export ships with a single, un-weighted `Assignments` group (weight 0). Set up weighted groups to match your section and your school's policy before the term opens; the scheme below matches the student syllabus default.

Suggested assignment groups and weights (mirrors the syllabus):

Assignment group	What goes in it	Suggested weight	Grading method
Practice & Checks	Independent Practices, Mid-Module Checks	40%	Auto-graded, unlimited attempts
Summative Assessments	The 12 module Summative Assessments	40%	Auto-graded, 2 attempts
Reviews & VR	Summative Reviews, VR resources/projects	20%	Completion (rubric for VR projects)

Canvas setup, in order: Gradebook settings → enable *Weight final grade based on assignment groups* → create the three groups above and set weights → move each existing assignment into its group → mark Summative Reviews and VR items as *Complete/Incomplete* → set your late/missing policy to match the syllabus (no per-day penalty; end-of-quarter zero; 10% deduction in the final 14 days). Use the **module requirements / progress view** as your at-a-glance completion signal across the ungraded scaffolding.

7 · Rubrics for teacher-graded items

Most items in this course auto-grade (numeric / multiple-choice with feedback), so teacher-scored rubrics are needed only for the **VR projects** and any **alternative assignments** (graphic organizer, poster, student-made explainer) offered in the lesson maps. Use one consistent 4-criteria rubric across all of them so grading stays about 5 minutes per student:

Criterion	Full credit	Partial	Not yet
Mathematical accuracy	Math is correct and complete	Minor errors; core idea sound	Key errors or missing work
Reasoning & representation	Clear method; correct use of notation/graphs/models	Method mostly clear; some gaps	Method unclear or absent
Communication	Explains the thinking clearly	Some explanation	Little or no explanation
Effort & honesty	A sincere, complete attempt in the student's own work	Incomplete attempt	Insincere / not own work

Paste this into each VR project and alternative assignment as a **native Canvas Rubric** so it scores in SpeedGrader. Keep the four criteria identical module to module.

8 · Assessment & activity catalog

Type	Scored as	Recommended teacher move
Independent Practice	Auto, unlimited attempts	Don't hand-grade. Scan item analysis for the most-missed problem.
Mid-Module Check	Auto, unlimited attempts	Use as your reteach signal (and live warm-up).
Summative Review	Completion	Assign before the assessment; run a live review over it.
Summative Assessment	Auto, 2 attempts	Review scores; draft targeted feedback for low scorers (Ignite AI).
VR Lesson Resource	Completion	Debrief briefly; no submission to mark.
VR Project / Alternative assignment	Rubric (Section 7)	Grade with the shared 4-criteria rubric.

Type	Scored as	Recommended teacher move
Prior Knowledge Check	Diagnostic	Read as a readiness signal at module start; don't weight heavily.

9 · Coach, not catch

- Feedback names the next step, not just the error - lead with what's working.
- Unlimited practice attempts are a learning tool; encourage retries rather than treating a low first attempt as a verdict.
- Use item analysis and Ignite AI to *find students who need help early*, then reach out - not to build a case.
- On suspected AI misuse or copied work, open a conversation: ask the student to talk through a problem. Treat it as formation.
- Public comments and live reviews model good mathematical thinking; they never single out weak work.

10 · Weekly announcement scripts

In an async section the weekly announcement is your main presence. Clone this template each week:

> **Subject: [Module N] - [this week's focus]**
> Hi mathematicians! This week we're in **[lesson/topic]**. Aim to finish **[the mini-lessons + practice due]** by **[date]**. The one thing not to miss: **[tricky concept / upcoming Summative]**. Stuck? Bring it to Study Hall on **[day/time]** - and remember every practice lets you retry with hints. You've got this. - [Your name]

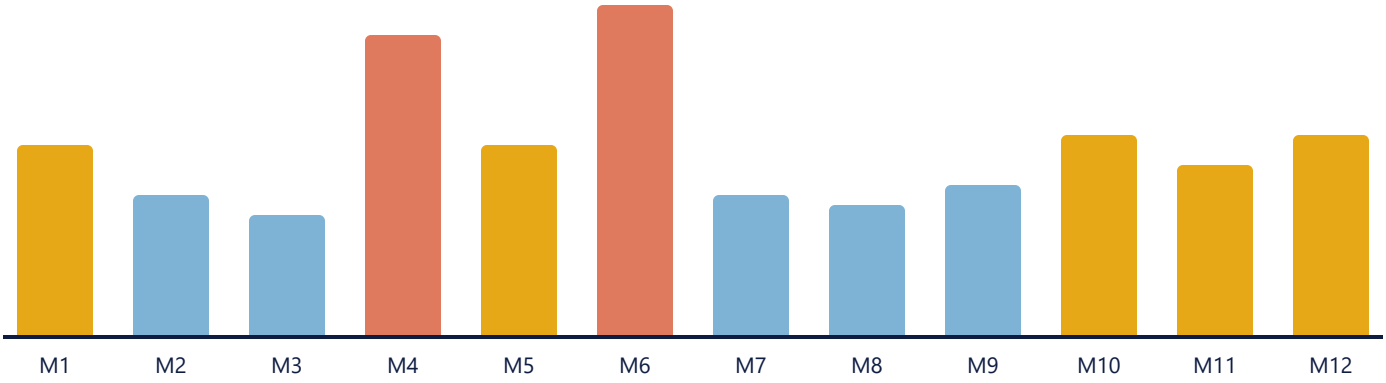
Situational scripts

- **Mid-module check-in:** "If the Mid-Module Check felt shaky, that's exactly what it's for - review the flagged concept and retry the practice before the summative."
- **Pre-assessment:** "The Module [N] Summative opens [date]. Do the Summative Review first - it mirrors the assessment. You get two attempts."
- **Re-engagement:** "Haven't seen your work in a bit - no penalty for a late day, but let's not let the quarter close on it. Reply here and I'll help you map a plan."

11 · Teacher workflow & workload map

A normal week, in order: scan Mid-Module Check / item analysis for the most-missed concept → grade any VR projects or alternative assignments with the shared rubric → review the week's Summative Assessment scores and draft targeted feedback → post the weekly announcement → answer messages and host Study Hall. Auto-graded practice costs ~0 hours; your time goes to feedback and outreach.

Heavier modules: Modules 4 and 6 are the densest (Module 6 has five lessons and the most pages; Module 4 is assignment-heavy). Budget extra time there and add a review week in a live section.



Relative facilitator load by module (estimated from lesson/assignment counts). Blue = light, gold = moderate, coral = heavy. Plan extra time on Modules 4 and 6.

12 · Standards coverage map

The course is aligned to the **Florida B.E.S.T. Standards for Mathematics, Grade 8**. The codes and benchmark text below are read directly from the course's own module lesson maps - these are the *primary performance standards* each module teaches and assesses. Keep this map in the teacher copy; standard codes do not appear on student pages.

By module (quick reference)

Module	Title	Semester	Primary B.E.S.T. standards
1	Laws of Exponents & Scientific Notation	1 (Q1)	MA.8.NSO.1.3, .1.4, .1.5, .1.6
2	Real Numbers	1 (Q1)	MA.8.NSO.1.1, .1.2, .1.7
3	Algebraic Expressions	1 (Q1)	MA.8.AR.1.1, .1.2, .1.3
4	Algebraic Equations & Inequalities in One Variable	1 (Q2)	MA.8.AR.2.1, .2.2, .2.3
5	Functions & Proportional Relationships	1 (Q2)	MA.8.F.1.1, .1.2; MA.8.AR.3.1
6	Linear Functions	1 (Q2)	MA.8.AR.3.2, .3.3, .3.4, .3.5
7	Systems of Equations	2 (Q3)	MA.8.AR.4.1, .4.2, .4.3
8	Bivariate Data	2 (Q3)	MA.8.DP.1.1, .1.2, .1.3
9	Probability & Simulated Experiments	2 (Q3)	MA.8.DP.2.1, .2.2, .2.3
10	Angle Relationships	2 (Q4)	MA.8.GR.1.4, .1.5, .1.6

Module	Title	Semester	Primary B.E.S.T. standards
11	Triangles & Their Side Lengths	2 (Q4)	MA.8.GR.1.1, .1.2, .1.3
12	Transformations & Similar Triangles	2 (Q4)	MA.8.GR.2.1, .2.2, .2.3, .2.4

By standard (with benchmark text)

Standard	Benchmark	Taught in
MA.8.NSO.1.1	Extend understanding of rational numbers to define irrational numbers within the real number system; locate an approximate value on a number line.	M2
MA.8.NSO.1.2	Plot, order, and compare rational and irrational numbers in various forms.	M2 (also M9, M11)
MA.8.NSO.1.3	Extend the Laws of Exponents to integer exponents; apply them to evaluate and generate equivalent numerical expressions (rational bases).	M1 (also M3)
MA.8.NSO.1.4	Express numbers in scientific notation to represent/approximate very large or small quantities; compare magnitudes.	M1
MA.8.NSO.1.5	Add, subtract, multiply, and divide numbers in scientific notation with procedural fluency.	M1
MA.8.NSO.1.6	Solve real-world problems involving operations with numbers in scientific notation.	M1
MA.8.NSO.1.7	Solve multi-step problems involving order of operations with rational numbers, including exponents and radicals.	M2 (also M11)
MA.8.AR.1.1	Apply the Laws of Exponents to generate equivalent algebraic expressions (integer exponents, monomial bases).	M3
MA.8.AR.1.2	Apply properties of operations to multiply two linear expressions with rational coefficients.	M3
MA.8.AR.1.3	Rewrite the sum of two expressions with a common monomial factor as a factor times a sum.	M3
MA.8.AR.2.1	Solve multi-step linear equations in one variable, rational coefficients, variables on both sides.	M4 (recurs in M5-M7, M10)
MA.8.AR.2.2	Solve two-step linear inequalities in one variable; represent solutions algebraically and graphically.	M4

Standard	Benchmark	Taught in
MA.8.AR.2.3	Given $x^2=p$ and $x^3=q$, determine the real solutions.	M4
MA.8.AR.3.1	Determine if a linear relationship is also a proportional relationship.	M5
MA.8.AR.3.2	Given a table, graph, or description of a linear relationship, determine the slope.	M6
MA.8.AR.3.3	Write an equation in slope-intercept form from a table, graph, or description.	M6
MA.8.AR.3.4	Graph a two-variable linear equation from a description, table, or slope-intercept equation.	M6
MA.8.AR.3.5	Determine and interpret slope and y-intercept in a real-world linear context.	M6
MA.8.AR.4.1	Determine which ordered pairs satisfy a system of two linear equations.	M7
MA.8.AR.4.2	From a graph of two linear equations, determine one / no / infinitely many solutions.	M7
MA.8.AR.4.3	Solve systems of two linear equations by graphing in a real-world context.	M7
MA.8.F.1.1	Determine whether a relation (pairs, table, graph, mapping) is a function; identify domain and range.	M5
MA.8.F.1.2	Determine whether a function (graph, equation, or table) is linear.	M5
MA.8.DP.1.1	Construct a scatter plot or line graph as appropriate for real-world bivariate data.	M8
MA.8.DP.1.2	Describe patterns of association in a scatter plot within context.	M8
MA.8.DP.1.3	Informally fit a straight line to a scatter plot with linear association.	M8 (introduced in M5-M6)
MA.8.DP.2.1	Determine the sample space for a repeated experiment.	M9
MA.8.DP.2.2	Find the theoretical probability of an event in a repeated experiment.	M9
MA.8.DP.2.3	Solve real-world probability problems and make predictions from theoretical probability.	M9

Standard	Benchmark	Taught in
MA.8.GR.1.1	Apply the Pythagorean Theorem to unknown side lengths in right triangles.	M11 (also M2, M4)
MA.8.GR.1.2	Apply the Pythagorean Theorem to distance between two points in the coordinate plane.	M11
MA.8.GR.1.3	Use the Triangle Inequality Theorem and the Converse of the Pythagorean Theorem.	M11
MA.8.GR.1.4	Solve problems with supplementary, complementary, vertical, or adjacent angles.	M10
MA.8.GR.1.5	Solve problems with interior and exterior angles of a triangle.	M10
MA.8.GR.1.6	Develop and use formulas for interior-angle sums of regular polygons.	M10
MA.8.GR.2.1	Identify the single transformation relating a preimage and image.	M12
MA.8.GR.2.2	Identify the scale factor of a single dilation.	M12
MA.8.GR.2.3	Describe/apply the effect of a single transformation using coordinates.	M12
MA.8.GR.2.4	Solve problems involving proportional relationships between similar triangles.	M12

> The course also threads earlier and later standards through its lessons as *Previous / Horizontal / Next* connections (e.g., MA.7 review at the start, MA.912 algebra links at the end). Those alignment notes live in each module's lesson-map document; the table above lists only the primary benchmarks each module is responsible for teaching and assessing.

13 · What makes this course classical & cultivating culture

Classical moves in this course. Pre-Algebra earns its place in a classical education by treating mathematics as reasoning, not recipe. The build supports this: students meet the *why* before the *how* (expanded form before the Laws of Exponents), extend the number system with real intellectual stakes (irrational numbers, MA.8.NSO.1.1), and connect procedures to real-world modeling (scientific notation, linear functions, similar triangles). The essential questions in the lesson maps - "What is the expanded form of an expression, and how does it help us write the Laws?" - are invitations to wonder, not just to compute.

Cultivating culture asynchronously. Name good mathematical thinking publicly (a clever approach in Study Hall, an elegant solution). Let your weekly announcement model curiosity, not just deadlines. Hold perseverance - one of the seven virtues - visible by celebrating the retry, since every practice item is built for it. Treat feedback as relationship: the student who struggled and came back is exactly who this course is for.

14 · Sensitive content

This is a mathematics course with **no mature themes, contested texts, or sensitive topics** to flag for families. The only advance notice worth sending is logistical: this is a VR school, so let families know a charged headset is needed for the immersive activities, and that an alternative assignment exists any time a student cannot get into VR. A short line in your welcome message covers it.